

INTERNSHIP REPORT

**Project title: Cancer Genomics Mutational Signature Comparison
Across Two Cancer Types (Lung Adenocarcinoma (LUAD)
and Colorectal Adenocarcinoma (COAD) Using Public TCGA
Data**

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Introduction to the Project

1.1 Cancer

Cancer is a disease characterized by the uncontrolled growth and division of abnormal cells due to the accumulation of genetic alterations. These alterations disrupt normal cellular functions, allowing cells to proliferate uncontrollably, invade surrounding tissues, and sometimes spread to other parts of the body through metastasis. Lung and colorectal cancers are among the leading causes of cancer-related deaths worldwide, making them important subjects for genomic research.

1.2 Lung Adenocarcinoma (LUAD) and Colon Adenocarcinoma (COAD)

Lung Adenocarcinoma (LUAD) is the most common subtype of non-small cell lung cancer and is often associated with tobacco smoking, environmental exposure, and genetic factors. Colon Adenocarcinoma (COAD) is the most common type of colorectal cancer and develops due to the accumulation of genetic mutations, lifestyle factors, aging, and defects in DNA repair mechanisms. Comparing these two cancers helps identify both shared and unique mutational processes involved in cancer development.

1.3 Mutational Signatures

Mutational signatures are characteristic patterns of DNA mutations generated by different biological and environmental processes during cancer development. These signatures are identified by analysing the types and sequence context of somatic mutations. The Catalogue of Somatic Mutations in Cancer (COSMIC) provides a comprehensive collection of known mutational signatures, such as **SBS1** (age-related mutations), **SBS4** (tobacco smoking), and **SBS2/SBS13** (APOBEC enzyme activity). Identifying these signatures helps researchers understand the mechanisms driving different cancers.

1.4 Cancer Genomics

Cancer genomics is the study of genetic alterations in cancer cells using high-throughput sequencing technologies and computational analysis. Public databases such as **The Cancer Genome Atlas (TCGA)** provide large-scale genomic datasets that enable researchers to investigate cancer-associated mutations. In this project, somatic mutation data from TCGA were analysed

using **SigProfilerMatrixGenerator** and **SigProfilerExtractor** to identify mutational signatures in Lung Adenocarcinoma (LUAD) and Colon

Adenocarcinoma (COAD). The extracted signatures were then compared with COSMIC reference signatures to identify shared and distinct mutational processes between the two cancer types.

Objectives

Most studies on mutational signatures investigate either individual cancer types or large pan-cancer datasets. However, there is limited work comparing the mutational signature profiles of Lung Adenocarcinoma (LUAD) and Colon Adenocarcinoma (COAD) using a reproducible computational approach. Such a comparison is important for identifying mutational processes that are common to both cancers as well as those that are unique, thereby providing insights into their underlying biological mechanisms.

In this mini project I wish to:

- 1.Download somatic mutation MAF files for LUAD and COAD from the TCGA GDC portal
- 2.Process and explore mutation data using maftools in R.
- 3.Run mutational signature extraction using SigProfilerExtractor on both cohorts.
- 4.Match extracted signatures to COSMIC SBS reference signatures.
- 5.Compare SBS profiles between the two cancer types and identify shared vs unique signatures.
- 6.Interpret findings biologically - e.g. SBS4 linked to tobacco in lung, SBS1 linked to age-related deamination.

Dataset Description

The datasets used in this project were obtained from **The Cancer Genome Atlas (TCGA)** through the **Genomic Data Commons (GDC) Data Portal** - a publicly available repository that contains comprehensive genomic, transcriptomic, and clinical data for various human cancers.

For this project, somatic mutation data from **Lung Adenocarcinoma (LUAD)** and **Colon Adenocarcinoma (COAD)** cohorts were downloaded.

The downloaded datasets were provided in **Mutation Annotation Format (MAF)**, which is a tab-delimited text file that stores annotated somatic mutation information for multiple cancer samples.

Each row in a MAF file represents a single somatic mutation identified in a patient sample, while each column contains detailed information describing the mutation.

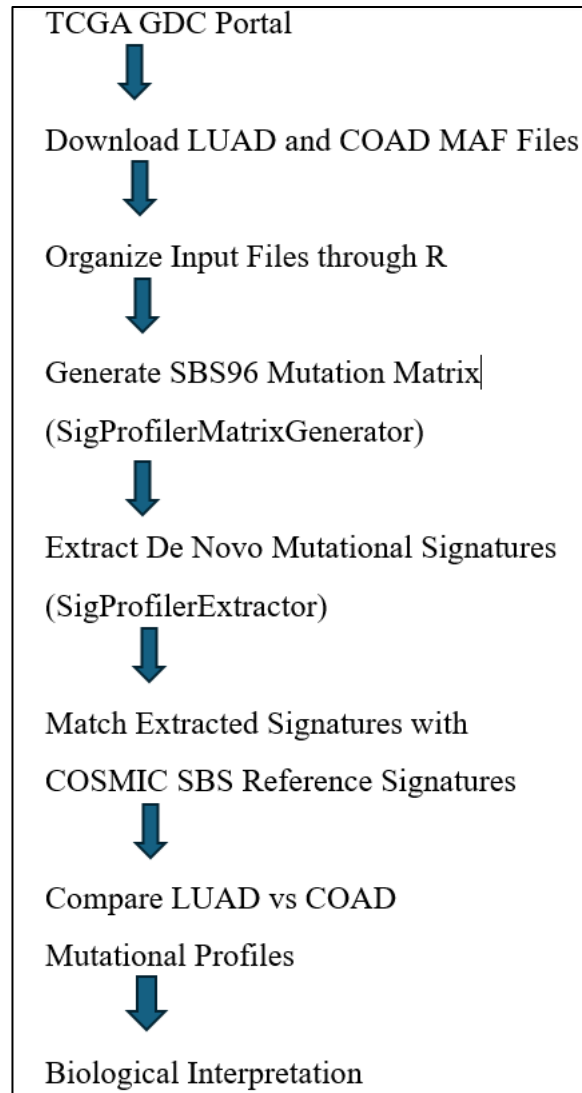
The LUAD dataset contained somatic mutation data from **lung adenocarcinoma patients**, whereas the COAD dataset contained mutations identified in **colon adenocarcinoma patients**. These datasets served as the primary input for mutational signature analysis using **SigProfilerMatrixGenerator** and **SigProfilerExtractor**.

The MAF files were used to generate **Single Base Substitution (SBS96) mutation matrices** using SigProfilerMatrixGenerator. These matrices were subsequently analysed using SigProfilerExtractor to identify de novo mutational signatures, which were then compared with the COSMIC reference signatures for biological interpretation.

Dataset	Cancer type	Source	File format	Reference Genome	Purpose
LUAD	Lung Adenocarcinoma	TCGA-GDC	MAF (.maf.gz)	GRCh38	Mutational Signature analysis
COAD	Colon Adenocarcinoma	TCGA-GDC	MAF (.maf.gz)	GRCh38	Mutational signature analysis

Methodology

The methodology adopted in this project involved downloading somatic mutation datasets from the TCGA Genomic Data Commons (GDC) portal, generating mutational matrices, extracting de novo mutational signatures, comparing them with the COSMIC reference database, and interpreting the identified signatures biologically. The workflow is illustrated below.



Following were the commands used in this project beginning from installing R, maftools, BiocManager, using SigProfiler tools, etc

Step No.	Task	Command(s)	Description / Output
1.	Update system	<code>sudo apt update</code>	Updates the package list.
2.	Install R base	<code>sudo apt install r-base -y</code>	Installs R and required dependencies.
3.	Start R	<code>R</code>	Starts R console. You should see ">".
4.	Install BiocManager	<code>install.packages("BiocManager")</code>	Installs Bioconductor package manager.
5.	Install maftools	<code>BiocManager::install("maftools")</code>	Installs maftools package.
6.	Load maftools	<code>library(maftools)</code>	Loads maftools library in R session.
7.	Read LUAD MAF file	<code>luad <- read.maf("LUAD.maf")</code>	Reads LUAD MAF file into R object 'luad'.
8.	Quick summary	<code>getSampleSummary(luad)</code>	Provides number of samples, mutation counts and basic statistics.
9.	Gene summary	<code>getGeneSummary(luad)</code>	Shows gene name and mutation frequency.
10.	Plot 1: Mutation Summary	<code>plotmafSummary(maf = luad)</code>	Displays total mutations, mutation types and variant classifications.
11.	Plot 2: Oncoplot	<code>oncoplot(maf = luad, top = 10)</code>	Shows top 10 mutated genes.
12.	Plot 3: Frequently Mutated Genes	<code>geneSummary <- getGeneSummary(luad) head(geneSummary)</code>	Lists frequently mutated genes.
13.	Read COAD MAF file	<code>coad <- read.maf("COAD.maf")</code>	Reads COAD MAF file into R object 'coad'.
14.	Compare LUAD vs COAD	<code>mafCompare(m1 = luad, m2 = coad, m1Name = "LUAD", m2Name = "COAD")</code>	Compares mutation burden and other features between LUAD and COAD.
15.	Exit R	<code>q()</code>	Exits R console.
16.	Create conda environment	<code>conda create -n sigprofiler python=3.12</code>	Creates a Python environment for SigProfiler.
17.	Activate environment	<code>conda activate sigprofiler</code>	Activates the environment.
18.	Install SigProfiler tools	<code>pip install SigProfilerMatrixGenerator pip install SigProfilerExtractor</code>	Installs SigProfilerMatrixGenerator and SigProfilerExtractor.
19.	Install reference genome (GRCh38)	<code>python -c "from SigProfilerMatrixGenerator import install as genInstall; genInstall.install('GRCh38')"</code>	Downloads and installs GRCh38 reference genome.
20.	Generate SBS96 mutation matrix	<code>SigProfilerMatrixGeneratorFunc(project = "LUAD", # or "COAD" reference_genome = "GRCh38", path_to_input_files = "./input_maf/")</code>	Generates SBS96, SBS384, SBS1536 and other matrices from MAF files.
21.	Extract mutational signatures	<code>sig.sigProfilerExtractor(input_type = "matrix", input_data = "./LUAD_SBS96.txt", # or COAD_SBS96.txt output = "LUAD_Extractor_Output/", # or COAD_Extractor_Output/ reference_genome = "GRCh38", cosmic_version = "3.3", make_plots = TRUE, csv_sigProfilerMatrixGeneratorNone = FALSE, run_type = "single",) nmf_replicates = 8</code>	Performs de novo signature extraction using NMF and generates results (plots, activities, matrices).
22.	List output files	<code>ls -R LUAD_Extractor_Output/</code>	Lists all result files.
23.	Check log file	<code>tail -50 LUAD_Extractor_Output/*.out</code>	Checks execution progress and logs.
24.	Compare signatures & interpret	(Performed using output plots, activities & COSMIC decomposition results)	Identify shared vs unique signatures between LUAD and COAD and interpret biologically.

Tools Used

Software Used

The computational analysis in this project was performed using **Python 3.12** within a dedicated **Conda (Miniconda)** environment, which ensured proper package management and dependency control.

R was also installed to facilitate the use of bioinformatics packages for handling Mutation Annotation Format (MAF) files.

The **BiocManager** package was used to install the **maftools** package, which provides functions for reading, summarizing, and visualizing somatic mutation data. Additionally, **GNU Nano** was used as a text editor to create and modify Python scripts required for mutational signature extraction.

Bioinformatics Tools Used

Several bioinformatics tools were used to perform mutational signature analysis. **SigProfilerMatrixGenerator** was employed to generate Single Base Substitution (SBS96) mutation matrices from the LUAD and COAD MAF files based on the GRCh38 reference genome.

These mutation matrices were then analysed using **SigProfilerExtractor**, which applies Non-negative Matrix Factorization (NMF) to extract de novo mutational signatures and compare them with the **COSMIC (Catalogue of Somatic Mutations in Cancer)** reference signature database.

The **maftools** package was used for reading and exploring MAF files and obtaining summary statistics of the mutation datasets.

Datasets Used

The datasets used in this study were obtained from **The Cancer Genome Atlas (TCGA)** through the **Genomic Data Commons (GDC) Data Portal**.

Publicly available somatic mutation datasets for **Lung Adenocarcinoma (TCGA-LUAD)** and **Colon Adenocarcinoma (TCGA-COAD)** were downloaded in **Mutation Annotation Format (MAF)**. These files contain annotated somatic mutations identified from tumour samples and served as the primary input for mutational signature analysis. The analyses were carried out using the **GRCh38** human reference genome

Results

I. Oncoplot Analysis of LUAD and COAD

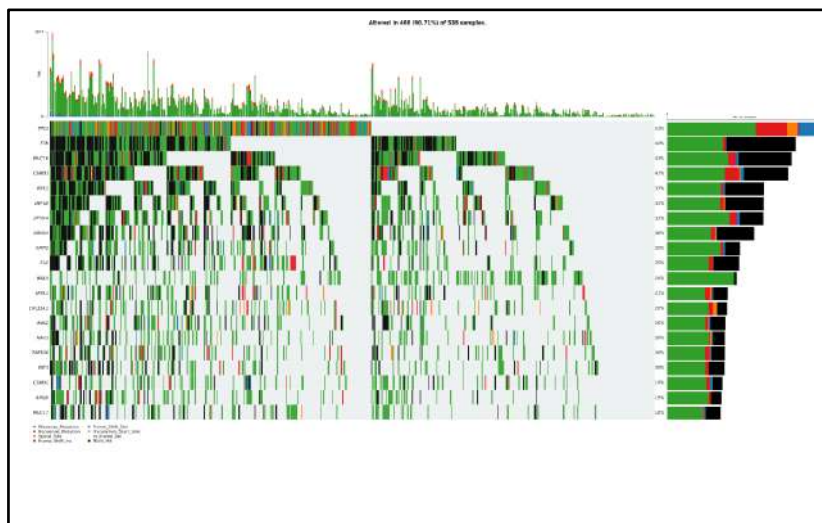
1. What is an Oncoplot?

An oncoplot is a graphical summary of somatic mutations across multiple tumour samples. Each column represents a tumour sample, each row represents a gene, colours indicate mutation types, the top bar plot shows mutation burden per sample, and the right-side bar plot shows the frequency of mutations in each gene.

2. Why is an Oncoplot Needed?

Oncoplots provide a quick overview of the genomic landscape of a cancer cohort, identify frequently mutated genes, reveal mutation patterns, and enable comparison between different cancer types.

3. LUAD Oncoplot



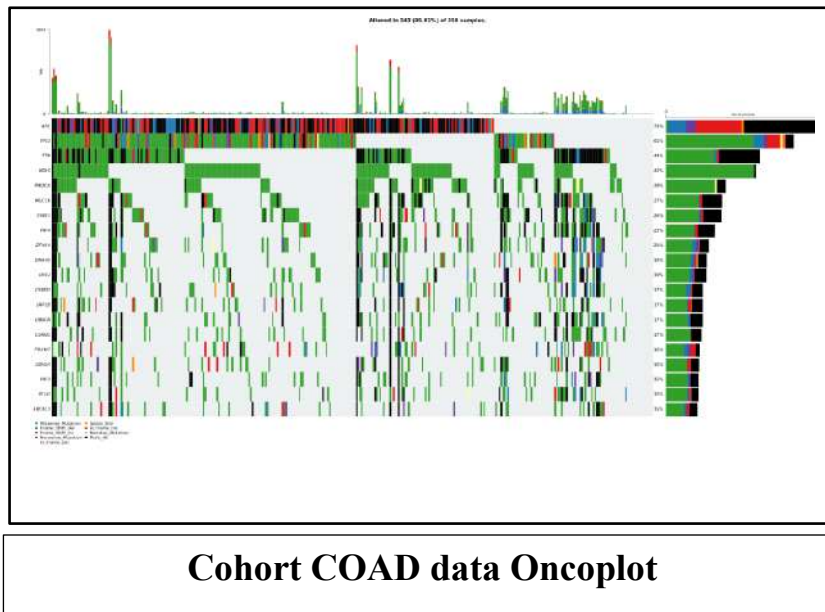
Cohort LUAD data Oncoplot

Interpretation

- 538 LUAD samples were analysed and 488 (90.71%) contained at least one genetic alteration.
- TP53 (53%) was the most frequently altered gene followed by TTN (44%) and MUC16 (43%).
- Other recurrently altered genes included CSMD3, RYR2, LRP1B and ZFHX4.

- Missense mutations predominated, indicating considerable genomic heterogeneity.
- Frequent TP53 alterations are consistent with its role as a major tumour suppressor in lung adenocarcinoma.

4. COAD Oncoplot



Interpretation

- 358 COAD samples were analysed and 343 (95.81%) showed at least one genetic alteration.
- APC (73%) was the most frequently altered gene followed by TP53 (61%), TTN (44%) and KRAS (42%).
- Other commonly altered genes included PIK3CA, MUC16, SYNE1 and FAT4.
- Missense mutations were the predominant mutation class.
- The prominence of APC and KRAS mutations reflects their established role in colorectal cancer development.

5. Comparative Observation

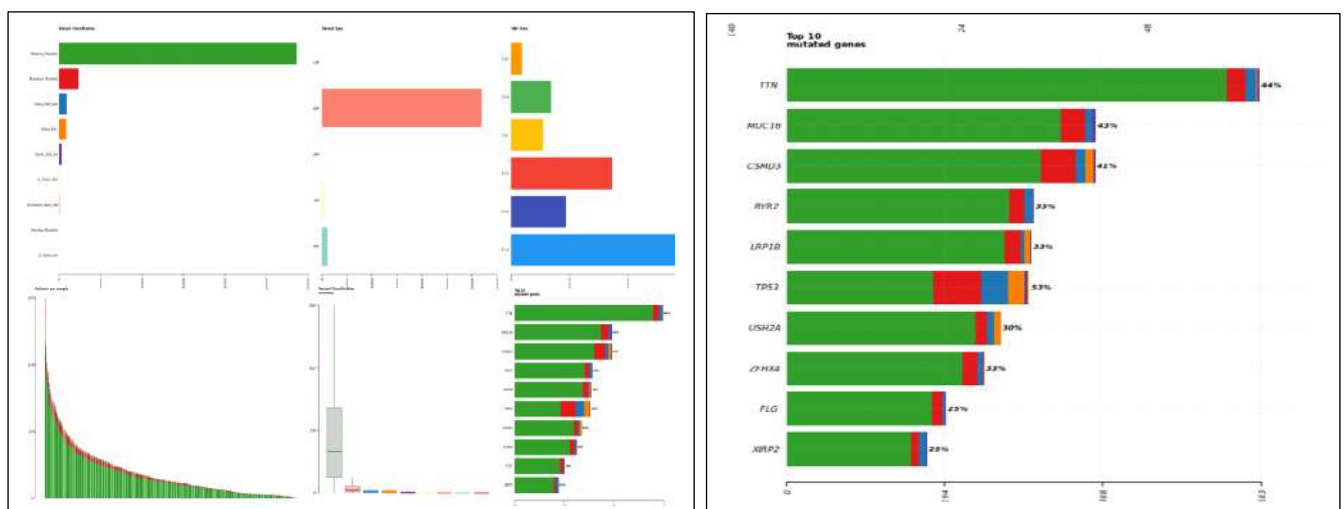
Both LUAD and COAD displayed high mutation frequencies, but the dominant driver genes differed. TP53 was commonly altered in both cancers, whereas APC and KRAS were characteristic of COAD. These differences highlight distinct molecular mechanisms underlying the two cancer types.

II. Top Mutated Genes Analysis of LUAD and COAD

1. Importance of Top Mutated Gene Analysis

The Top Mutated Genes plot summarizes the genes that are most frequently altered across all tumour samples in a cohort. It helps identify potential driver genes, understand the molecular characteristics of a cancer type, prioritize biomarkers for diagnosis and therapy, and compare the genetic landscape between different cancers.

2. LUAD Top Mutated Genes

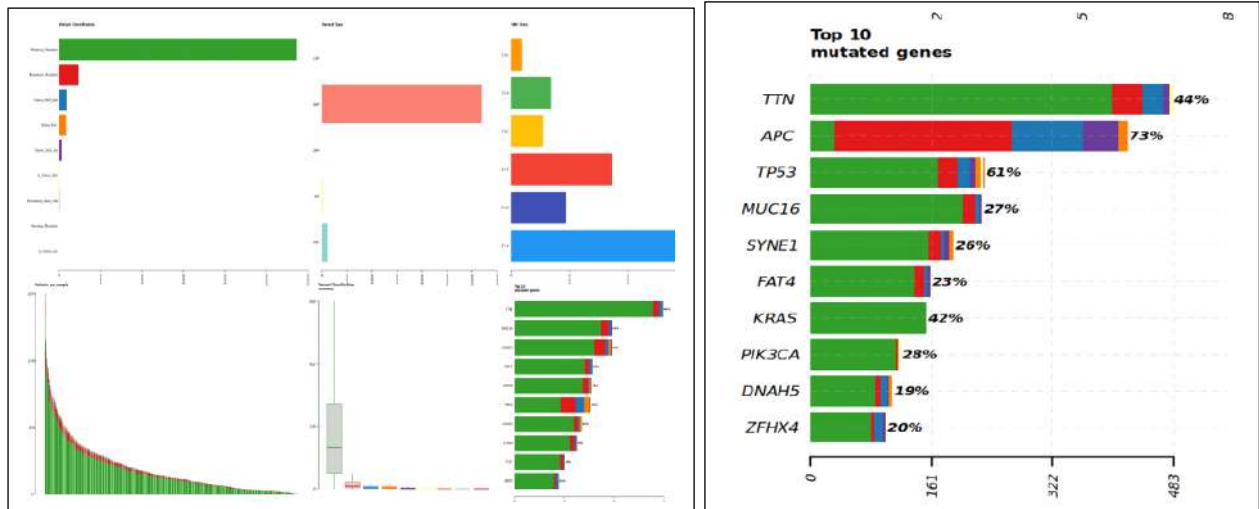


Summary of all plots and Top 10 mutated gene plots for LUAD

Interpretation:

- TTN (44%), MUC16 (43%) and CSMD3 (41%) were among the most frequently mutated genes.
- TP53 showed mutations in 53% of the samples, highlighting its major role in lung adenocarcinoma.
- Other recurrently mutated genes included RYR2 (33%), LRP1B (33%), USH2A (30%), ZFHX4 (33%), FLG (25%) and XIRP2 (25%).
- The predominance of TP53 mutations reflects disruption of cell-cycle regulation, while frequent TTN and MUC16 mutations indicate a high mutation burden typical of LUAD.

3. COAD Top Mutated Genes



Summary of all plots and Top 10 mutated gene plots for COAD

Interpretation:

- APC was the most frequently mutated gene (73%), followed by TP53 (61%), TTN (44%) and KRAS (42%).
- Other commonly mutated genes included PIK3CA (28%), MUC16 (27%), SYNE1 (26%), FAT4 (23%), DNAH5 (19%) and ZFHX4 (20%).
- Frequent APC and KRAS mutations are characteristic features of colorectal tumorigenesis, while TP53 mutations contribute to tumour progression.

Comparison of LUAD and COAD

Feature	LUAD	COAD
Most frequently mutated gene	TP53 (53%)	APC (73%)
Second major gene	TTN (44%)	TP53 (61%)
Other common genes	MUC16, CSMD3, RYR2, LRP1B	KRAS, PIK3CA, MUC16, SYNE1
Key biological implication	Predominant alterations in tumour suppressor pathways with high mutation burden.	Strong involvement of Wnt signalling (APC) and MAPK signalling (KRAS).
Shared genes	TP53, TTN, MUC16	TP53, TTN, MUC16

5. Overall Interpretation

The top mutated gene analysis demonstrates distinct molecular landscapes for LUAD and COAD. Both cancers share recurrent mutations in TP53, TTN and MUC16, suggesting common mechanisms of genomic instability. However, LUAD is characterized by frequent TP53, CSMD3 and LRP1B mutations, whereas COAD is dominated by APC and KRAS mutations, reflecting differences in the biological pathways driving tumour development. These findings complement the mutational signature analysis by identifying genes that are most frequently affected within each cancer type.

III. Mutational Signature Analysis

What are Mutational Signatures?

Mutational signatures are characteristic patterns of somatic mutations that arise due to different endogenous and exogenous mutational processes acting on the genome over time. Each signature represents a specific combination of the 96 possible single base substitution (SBS96) mutation contexts and can often be linked to particular biological mechanisms such as ageing, tobacco smoking, DNA mismatch repair deficiency, or APOBEC enzyme activity. Identification of these signatures helps reveal the underlying causes of cancer development and progression.

What is the SBS96 Decomposition Plot?

The SBS96 decomposition plot compares each **de novo mutational signature** extracted from the tumour samples with known **COSMIC SBS reference signatures**. The left panel shows the **original extracted signature** and the **reconstructed signature**, while the right panel displays the combination of COSMIC signatures that best explains the extracted profile along with their percentage contributions. Statistical measures such as cosine similarity, correlation, L1/L2 error and KL divergence indicate how accurately the reconstructed profile matches the original extracted signature.

Interpretation of LUAD Mutational Signatures

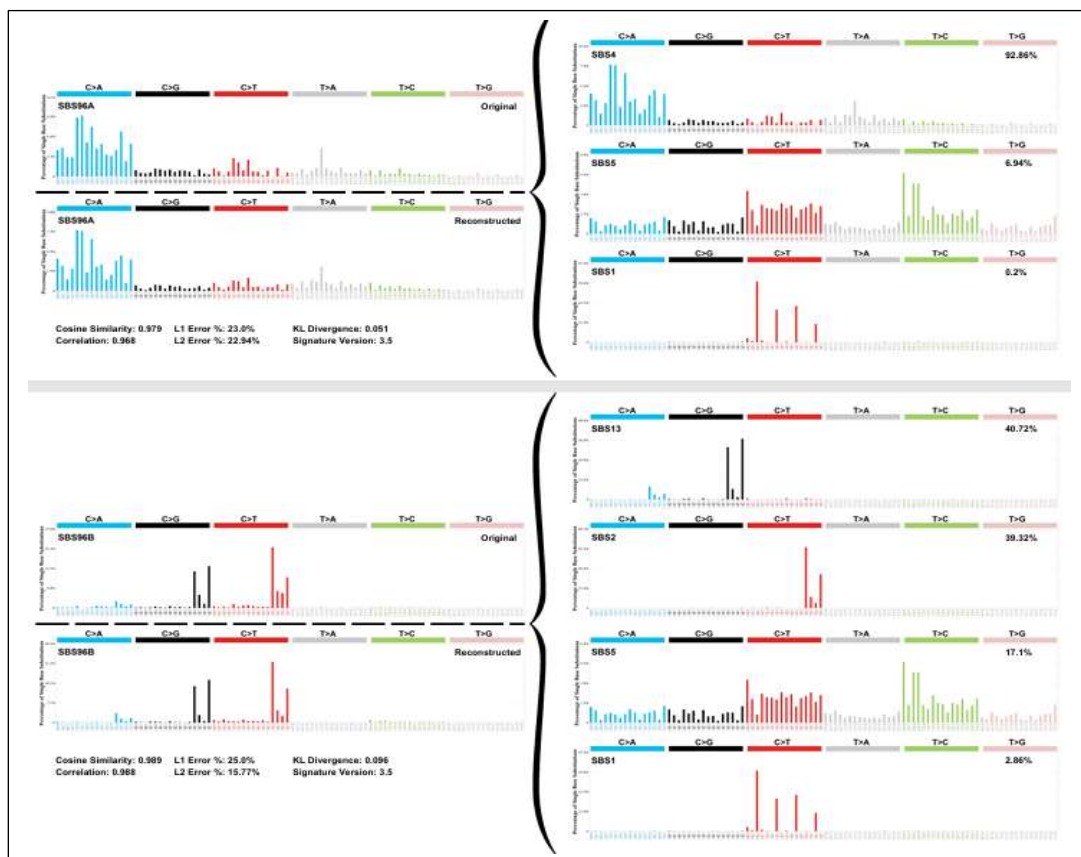
SigProfilerExtractor identified **three de novo mutational signatures** in the LUAD dataset, namely **SBS96A**, **SBS96B**, and **SBS96C**. These signatures showed high similarity with known COSMIC SBS signatures, indicating reliable decomposition into biologically relevant mutational processes.

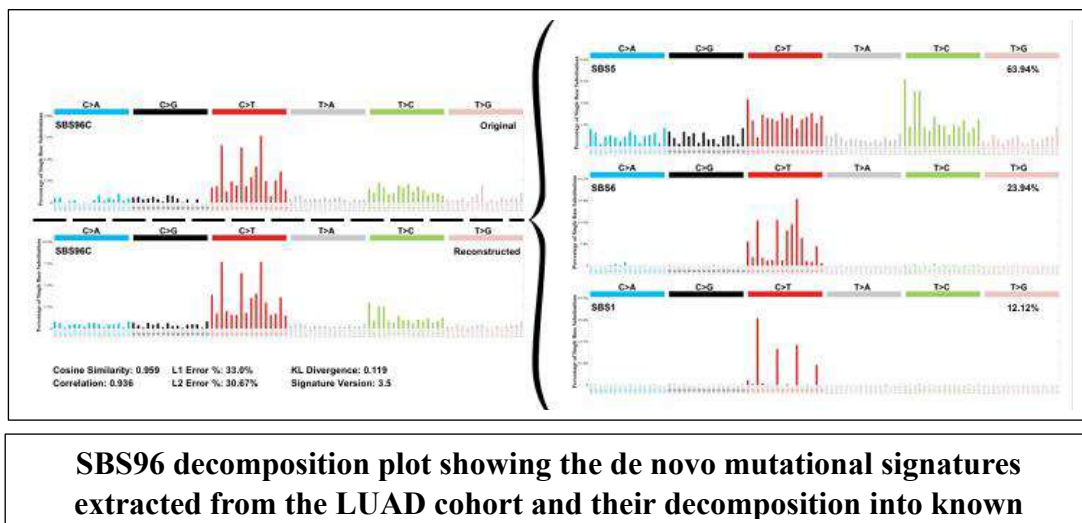
Signature SBS96A

SBS96A was primarily composed of:

- **SBS4 – 92.86%**
- **SBS5 – 6.94%**
- **SBS1 – 0.20%**

The cosine similarity between the extracted and reconstructed signatures was **0.979**, indicating an excellent match. SBS4 is a well-established signature associated with **tobacco smoking**, suggesting that tobacco-related mutagenesis is the dominant mutational process in this subset of LUAD samples. The small contributions from SBS5 and SBS1 represent background clock-like mutational processes.





What are COSMIC SBS96 Signature Plots?

LUAD:

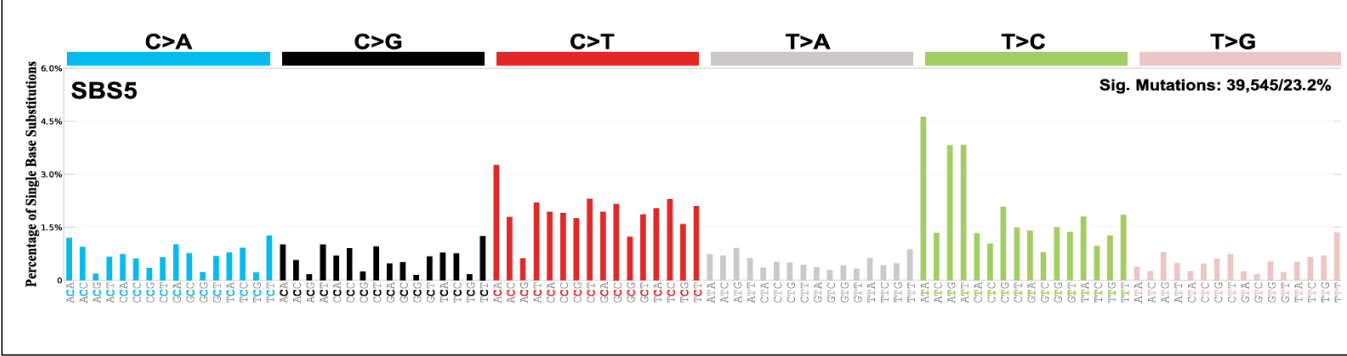
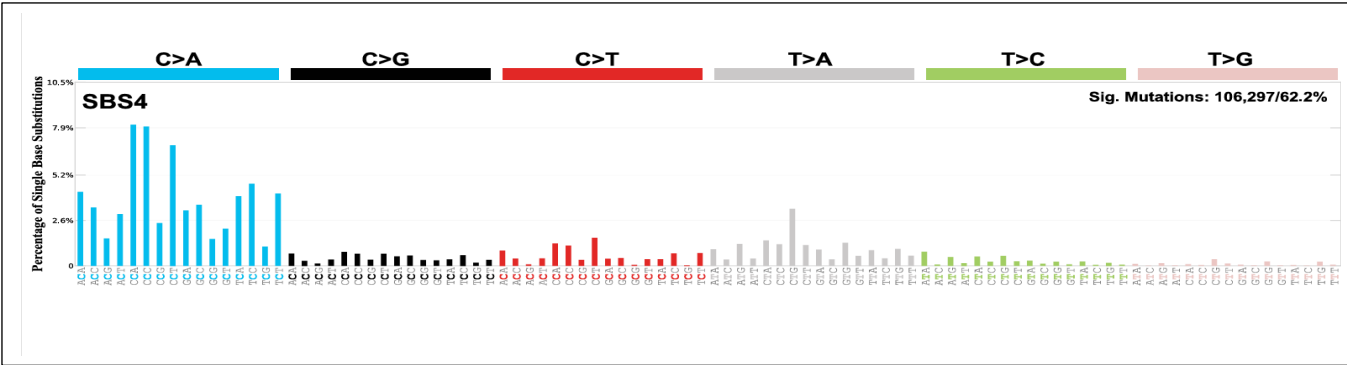
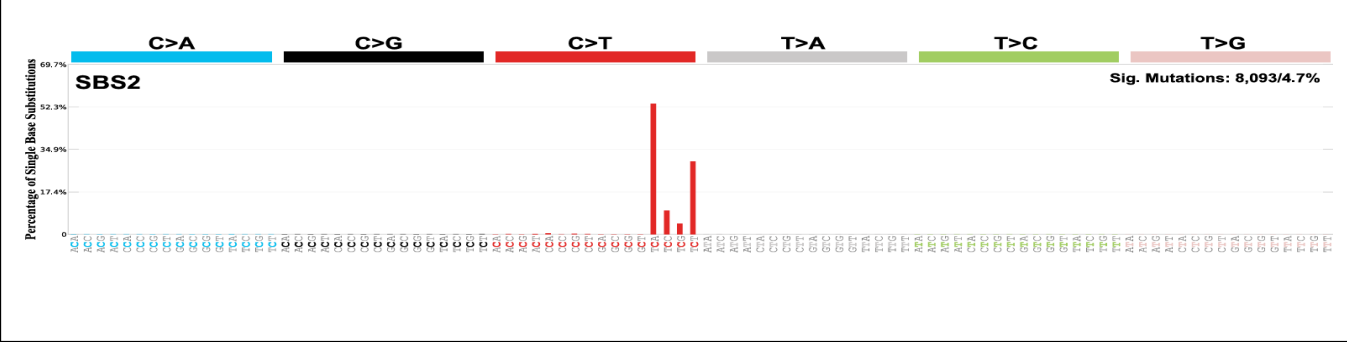
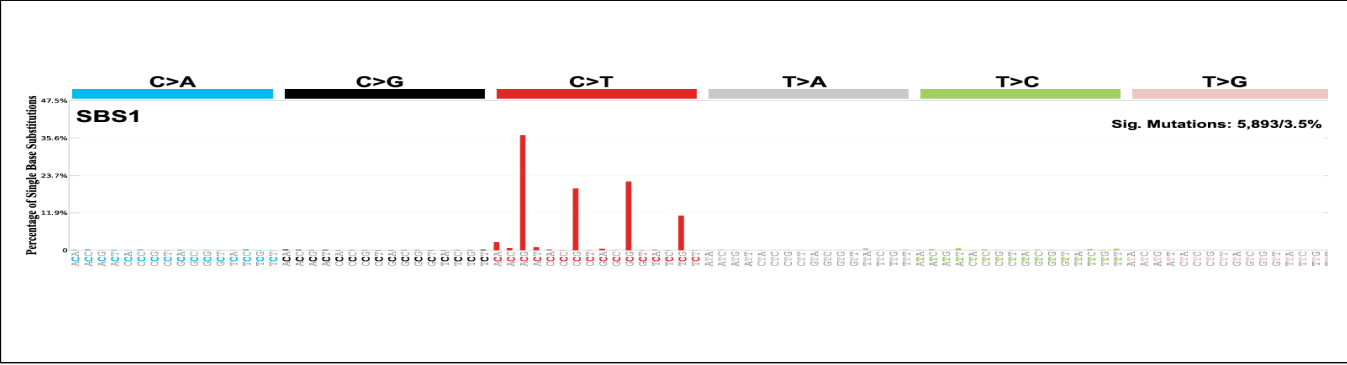
After matching the extracted mutational signatures to the COSMIC database, SigProfiler generates SBS96 signature plots for every COSMIC signature detected in the dataset. Each plot displays the characteristic pattern of mutations across the 96 trinucleotide mutation contexts. These plots help visualize the biological mutation process represented by each COSMIC signature and validate the signature assignment obtained during decomposition.

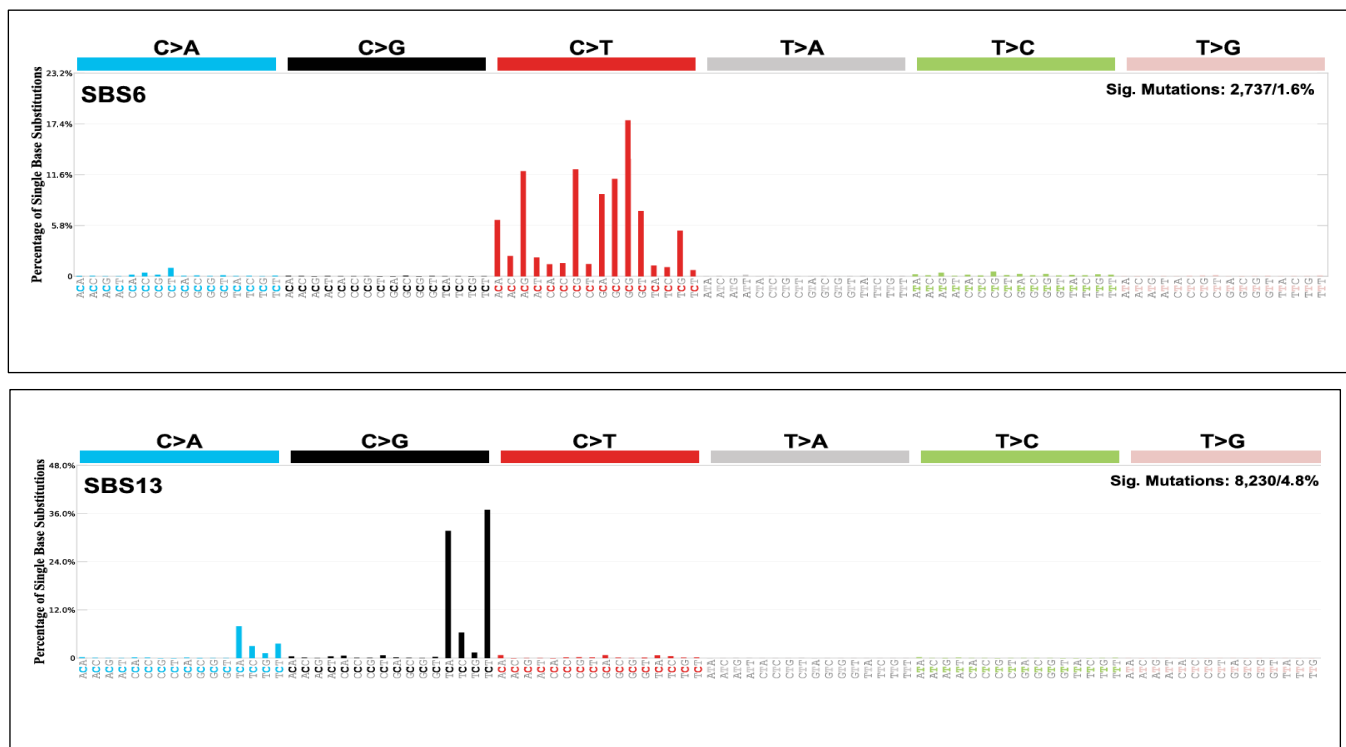
Each image represents **one COSMIC mutational signature**.

Each graph contains:

- **X-axis:** 96 mutation contexts (trinucleotide substitutions)
- **Y-axis:** Percentage contribution of each mutation type
- **Top coloured bars:** Six substitution classes
 - C>A
 - C>G
 - C>T
 - T>A
 - T>C
 - T>G

- **Sig. Mutations:** Total mutations assigned to that signature and its percentage contribution in the LUAD dataset.





Biological Interpretation of Each Signature

SBS1:

Biological process

- Age-related spontaneous deamination of methylated cytosine.

Observation

- Dominated by C>T substitutions.

Interpretation

- Indicates normal accumulation of mutations with ageing.
- Present at low frequency in LUAD (~3.5% of assigned mutations).

SBS2 :

Biological process

- APOBEC cytidine deaminase activity.

Observation

- Strong C>T mutation peaks.

Interpretation

- Indicates APOBEC-mediated DNA editing contributing to tumour evolution.
- Represents ~4.7% of mutations.

SBS4 :

Biological process

- Tobacco smoke exposure.

Observation

- Characteristic C>A substitutions dominate the profile.

Interpretation

- This is the dominant mutational process in LUAD.
- Accounts for approximately **62.2%** of all assigned mutations, highlighting smoking as the primary mutagenic factor.

SBS5 :

Biological process

- Clock-like endogenous mutational process.

Observation

- Broad distribution across all substitution types.

Interpretation

- Represents continuous mutation accumulation throughout life.
- Contributes approximately **23.2%** of mutations.

SBS6 :

Biological process

- DNA mismatch repair deficiency (MMR deficiency).

Observation

- High frequency of C>T substitutions.

Interpretation

- Suggests defective DNA repair in a subset of LUAD samples.
- Contributes about **1.6%** of mutations.

SBS13 :

Biological process

- APOBEC enzyme activity.

Observation

- Characteristic C>G mutation peaks.

Interpretation

- Indicates APOBEC-mediated mutagenesis occurring alongside SBS2.
- Accounts for approximately **4.8%** of mutations.

Interpretation of COAD Mutational Signatures

SigProfilerExtractor identified two de novo mutational signatures in the COAD dataset, namely SBS96A and SBS96B. These signatures were successfully decomposed into known COSMIC SBS signatures with high cosine similarity, indicating that the extracted signatures accurately represent biologically relevant mutational processes.

Signature SBS96A

The extracted signature **SBS96A** was decomposed into the following COSMIC signatures:

- **SBS4 – 67.68%**
- **SBS5 – 18.52%**
- **SBS13 – 6.90%**
- **SBS2 – 6.62%**
- **SBS1 – 0.28%**

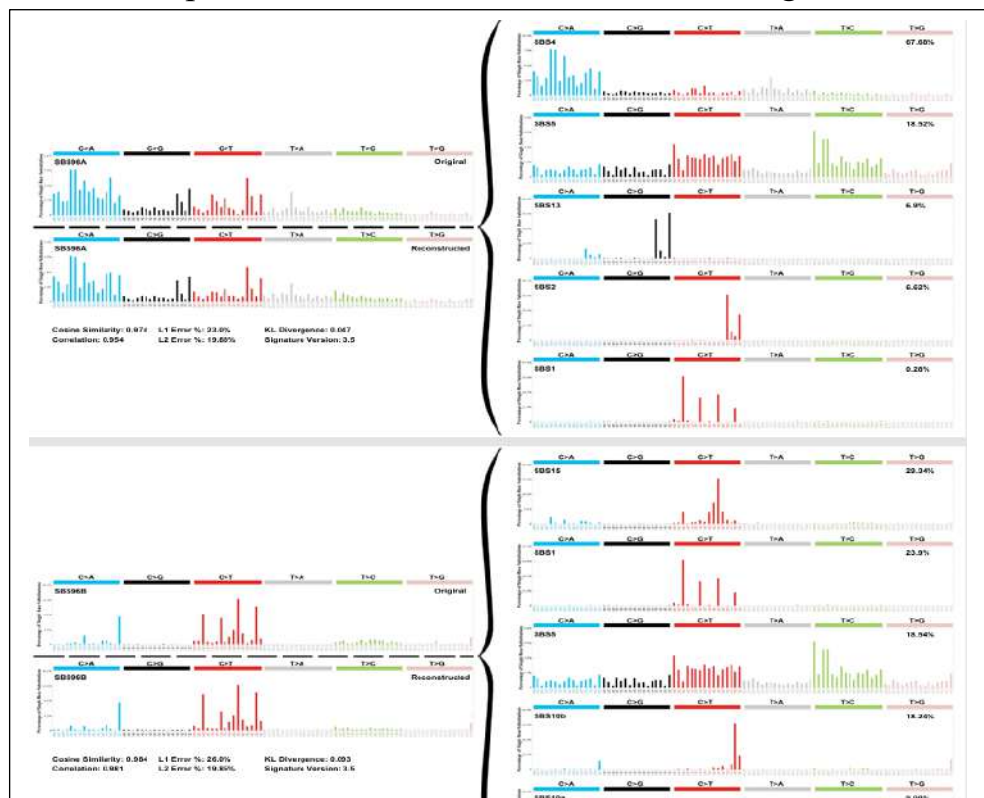
The reconstructed signature exhibited a **cosine similarity of 0.974** and a **correlation of 0.954**, indicating a strong agreement with the extracted signature.

The low KL divergence (0.047) further confirms the quality of the decomposition.

Biological Interpretation

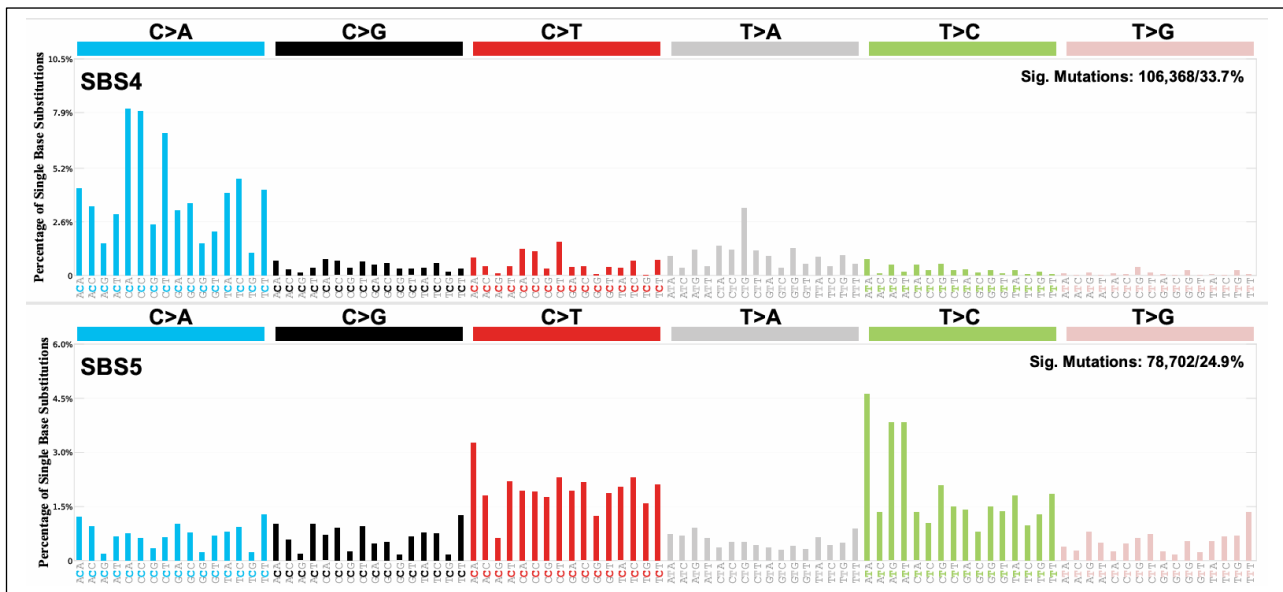
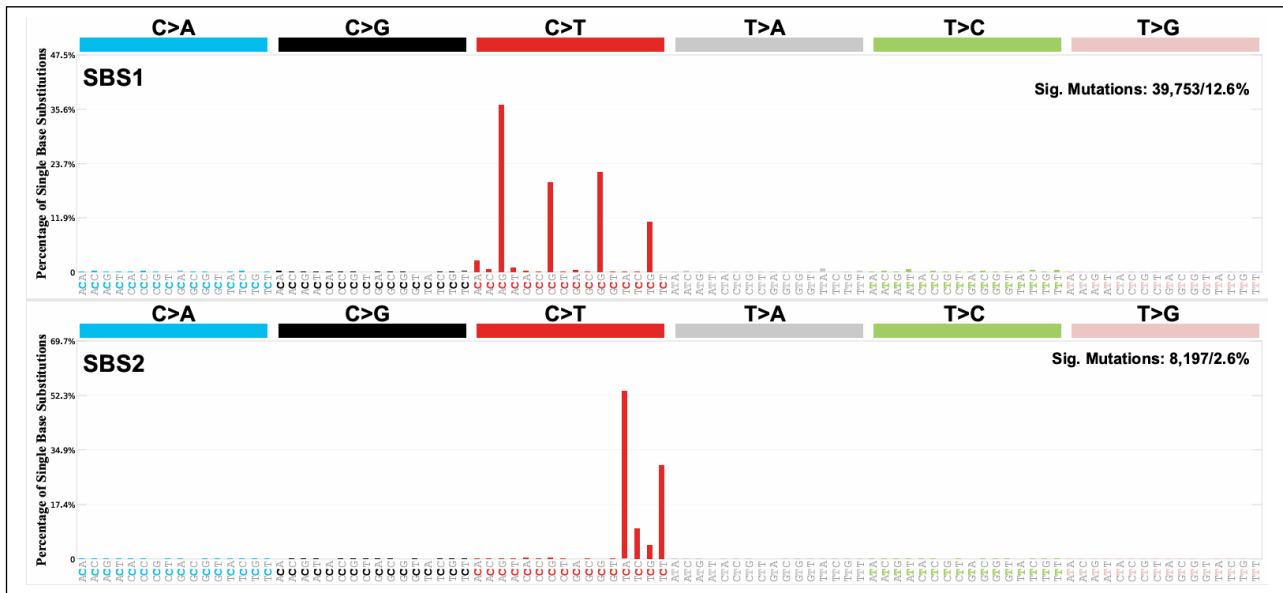
- **SBS4** is associated with **tobacco smoke exposure** and represents the largest contribution to this signature.
- **SBS5** is a clock-like signature that accumulates gradually with age and is commonly observed across many cancer types.
- **SBS2** and **SBS13** are associated with **APOBEC cytidine deaminase activity**, indicating cytosine deamination as an active mutational process.
- **SBS1** is another clock-like signature caused by spontaneous deamination of 5-methylcytosine and reflects normal age-related mutation accumulation.

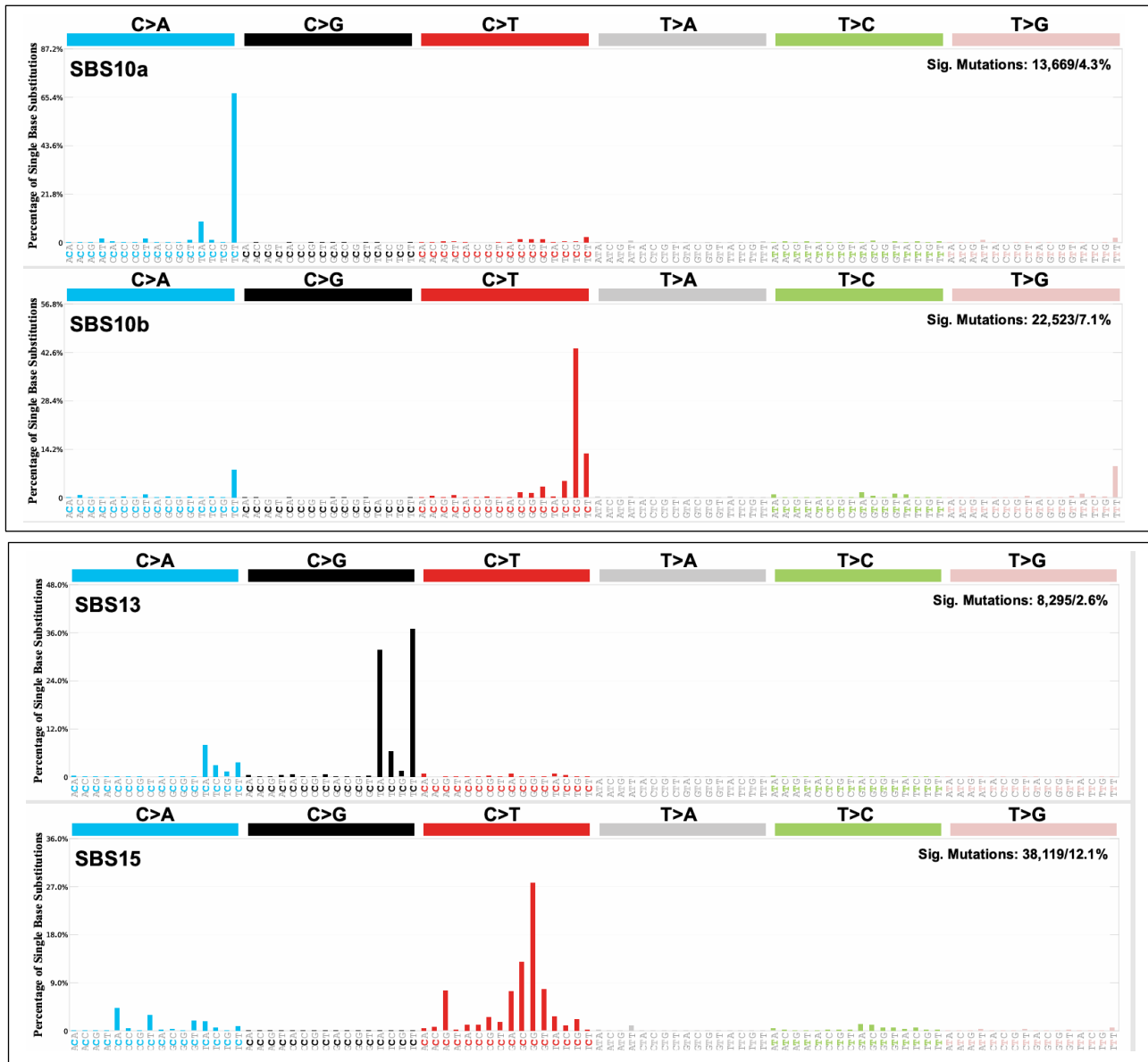
The predominance of SBS4 suggests that tobacco-related mutagenesis contributes substantially to this subset of COAD samples, while SBS5, SBS2 and SBS13 indicate the combined influence of endogenous ageing processes and APOBEC-mediated mutagenesis.



SBS96 decomposition plot showing the de novo mutational signatures extracted from the COAD cohort and their decomposition into known COSMIC SBS reference signatures using

COSMIC SBS96 Signature Plots for COAD :





Interpretation of Individual COSMIC Signatures

SBS1:

SBS1 is a **clock-like mutational signature** caused by the spontaneous deamination of 5-methylcytosine, producing predominantly **C>T substitutions at CpG sites**. It accumulates gradually with increasing age and is commonly observed in most human cancers.

Interpretation:

The presence of SBS1 indicates that normal age-related mutational processes have contributed to mutation accumulation in the COAD samples. Since this

signature is present in almost all cancers, it reflects endogenous cellular ageing rather than a cancer-specific mechanism.

SBS2 :

SBS2 is characterized mainly by **C>T and C>G mutations** and is associated with the activity of the **APOBEC family of cytidine deaminases**.

Interpretation:

Although present at a relatively low frequency, SBS2 suggests that APOBEC-mediated DNA editing has contributed to mutations in a subset of tumour cells. APOBEC activity increases genomic instability and may accelerate tumour evolution.

SBS4 :

SBS4 is characterized by a predominance of **C>A substitutions** and is the classical mutational signature associated with **tobacco smoke exposure**.

Interpretation:

SBS4 represents the largest mutational contribution in this dataset, accounting for approximately one-third of all detected mutations. This suggests that tobacco-associated DNA damage has played a major role in shaping the mutational landscape of this cohort. Although COAD is not classically considered a tobacco-associated cancer, smoking has been reported to increase colorectal cancer risk and may contribute to tumour development in certain patients.

SBS5 :

SBS5 is another **clock-like mutational signature** that accumulates steadily throughout life. Unlike SBS1, its exact biological cause remains unknown.

Interpretation:

The high contribution of SBS5 indicates that endogenous mutational processes occurring during normal ageing have substantially influenced mutation accumulation in the COAD samples. SBS5 is commonly detected in many cancer types.

SBS10a :

SBS10a is characterized by distinctive **C>A substitutions** and is associated with mutations affecting the proofreading activity of **DNA polymerase epsilon (POLE)**.

Interpretation:

The presence of SBS10a suggests that a subset of tumours may possess defective DNA replication proofreading, resulting in elevated mutation rates and increased genomic instability.

SBS10b:

SBS10b is another signature linked to **POLE proofreading deficiency**, characterized predominantly by **C>T substitutions**.

Interpretation:

Together with SBS10a, this signature supports the possibility of defective DNA polymerase proofreading in some COAD samples. POLE-mutant colorectal tumours are often hypermutated and possess distinct biological characteristics.

SBS13 :

SBS13 is also produced by **APOBEC enzyme activity** and frequently occurs alongside SBS2.

Interpretation:

The simultaneous presence of SBS2 and SBS13 confirms that APOBEC-mediated cytosine deamination contributes to mutagenesis in a subset of colorectal tumours, although its overall contribution remains relatively small.

SBS15 :

SBS15 is associated with **DNA mismatch repair (MMR) deficiency** and is commonly observed in **microsatellite instability (MSI)** colorectal cancers.

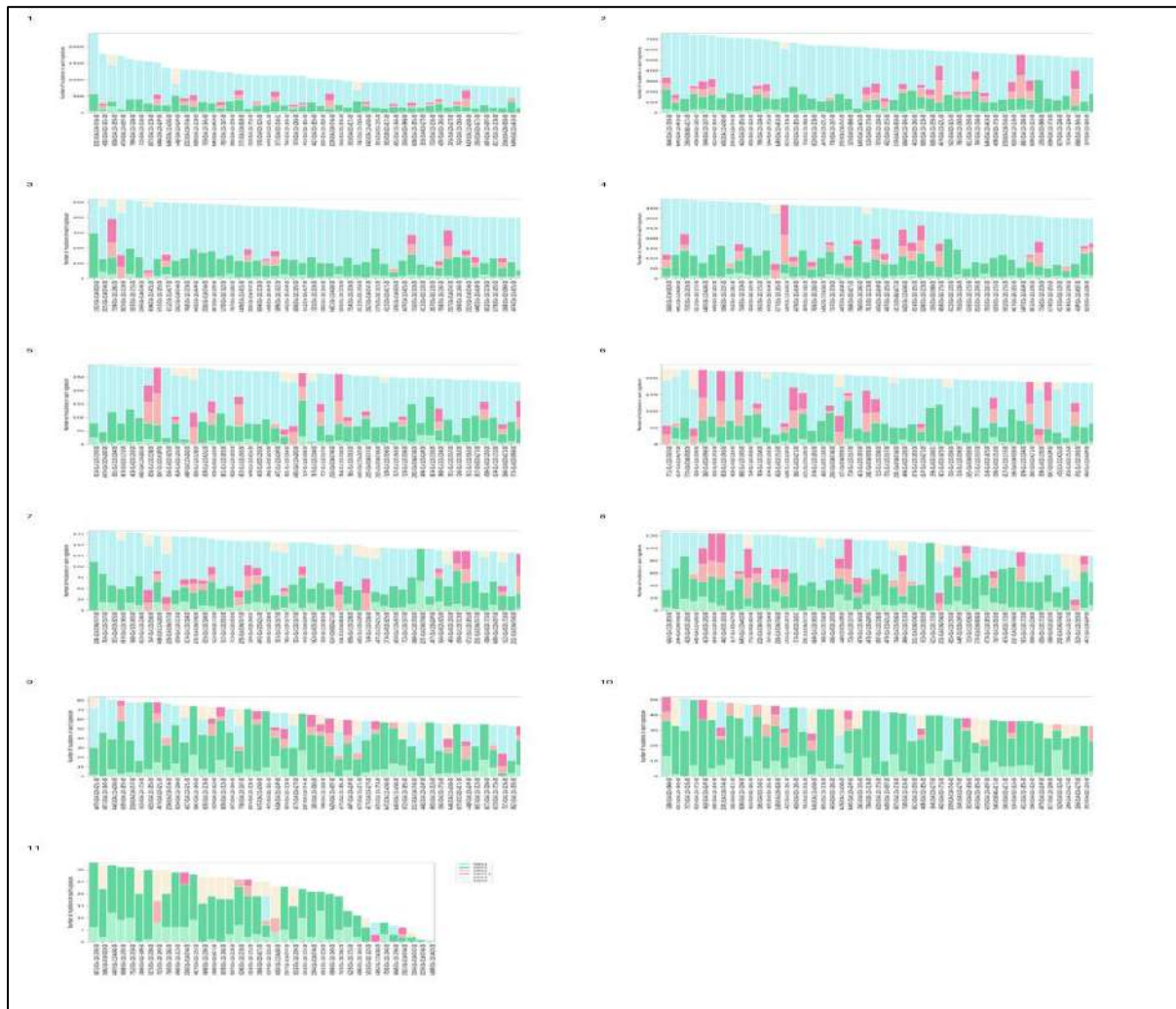
Interpretation:

The presence of SBS15 indicates that defective mismatch repair contributes significantly to mutation accumulation in the analysed COAD cohort. MMR deficiency is a well-established molecular feature of a subset of colorectal cancers and has important implications for prognosis and immunotherapy response.

Signature Activity Plot

It shows activity of each signature and contribution per sample. It compares the signatures that dominate LUAD and COAD.

LUAD:



The signature activity plot shows the contribution of each COSMIC mutational signature across individual LUAD samples. Each stacked bar represents one tumor sample, while different colors indicate different mutational signatures (SBS1, SBS2, SBS4, SBS5, SBS6 and SBS13). The height of each colored segment corresponds to the number of mutations attributed to that signature.

Interpretation:

SBS4 is the predominant signature in most LUAD samples, indicating that tobacco smoking is the major mutational process contributing to lung adenocarcinoma.

SBS5 is consistently present across nearly all samples, representing a clock-like mutational process that accumulates with age.

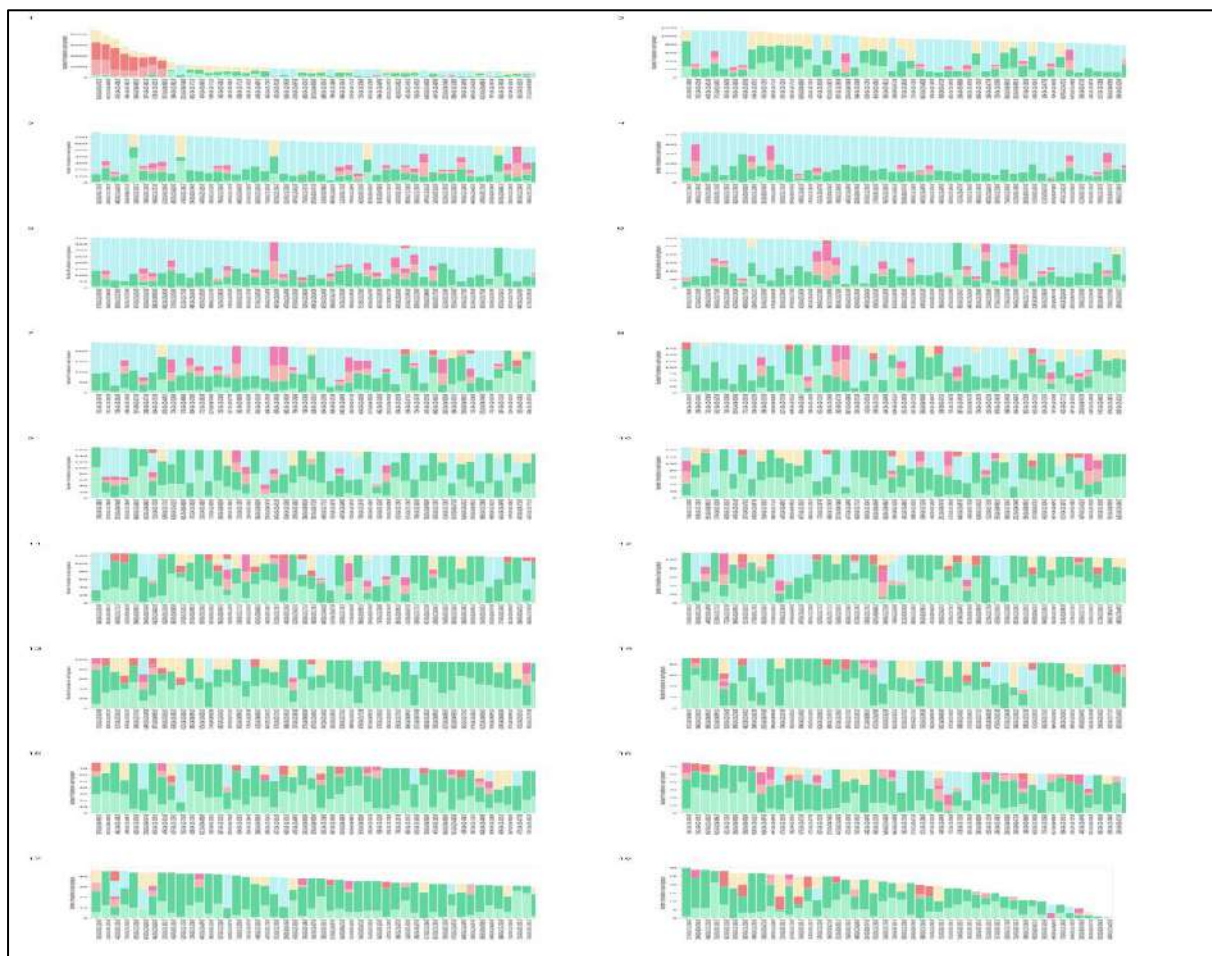
SBS1 is observed at lower levels in most samples and also reflects age-related spontaneous deamination of methylated cytosines.

SBS2 and SBS13 are present in a subset of samples, suggesting APOBEC cytidine deaminase activity contributes to mutagenesis in some tumors.

SBS6 appears only in a few samples and at relatively low activity, indicating that mismatch repair deficiency is uncommon in this LUAD cohort.

The activity of these signatures varies considerably among patients, demonstrating the heterogeneity of mutational processes in LUAD.

COAD:



The stacked bar plots represent the activity of different COSMIC mutational signatures across individual colorectal adenocarcinoma (COAD) samples. Each bar corresponds to one patient, and each color denotes a specific mutational signature.

Interpretation

- **SBS5** is the dominant signature in most COAD samples, indicating a ubiquitous clock-like mutational process associated with aging.
- **SBS1** is consistently present across many samples, reflecting spontaneous deamination of methylated cytosines and age-related mutation accumulation.
- A subset of samples exhibits increased **SBS6**, suggesting **DNA mismatch repair (MMR) deficiency** and **microsatellite instability (MSI)**, which are characteristic of a proportion of colorectal cancers.
- **SBS2** and **SBS13** are observed in some tumors, indicating **APOBEC-mediated cytidine deaminase activity** contributes to mutagenesis in those cases.
- Considerable variation in signature composition among patients demonstrates the **molecular heterogeneity of COAD**, where multiple endogenous and DNA repair-related mutational processes shape the tumor genome

Comparative Analysis of LUAD and COAD

The following section compares the major results obtained from mutation profiling and mutational signature analysis of Lung Adenocarcinoma (LUAD) and Colorectal Adenocarcinoma (COAD). The comparison highlights similarities and differences in mutation burden, driver genes and underlying mutational processes.

1. Mutation Summary Plot Comparison

Mutation summary plots generated using `maftools` (`plotmafSummary`) provide an overview of the mutational landscape, including variant classifications, variant types, SNV classes, variants per sample and frequently mutated genes.

LUAD exhibited a relatively higher mutation burden and greater inter-sample variability. Missense mutations were the predominant mutation class, with SNPs representing the major variant type. The SNV profile showed a substantial contribution of C>A substitutions, consistent with smoking-related DNA damage.

COAD also showed missense mutations as the predominant variant classification and SNPs as the major variant type. However, the SNV profile was dominated by C>T substitutions, reflecting endogenous mutational processes such as spontaneous cytosine deamination and age-related mutations.

Overall, LUAD demonstrated greater mutational heterogeneity, whereas COAD displayed a comparatively more uniform mutation profile.

2. Oncoplot Comparison

Oncoplots visualize the mutation status of the most frequently mutated genes across all samples and help identify recurrent driver mutations.

LUAD was characterized by frequent mutations in TTN, TP53, MUC16, CSMD3, RYR2 and LRP1B. Many of these genes are large structural genes commonly mutated in smoking-associated lung cancers.

In contrast, COAD was dominated by mutations in APC, TP53, KRAS and PIK3CA, which are well-established driver genes involved in colorectal tumorigenesis. APC mutations indicate dysregulation of the Wnt signalling pathway, whereas KRAS and PIK3CA mutations activate MAPK and PI3K signalling pathways.

These findings demonstrate distinct molecular mechanisms underlying the two cancer types.

3. Top Mutated Gene Comparison

The top mutated gene plots summarize the mutation frequency of the most frequently altered genes.

LUAD showed TTN, TP53 and MUC16 among the most frequently mutated genes, suggesting widespread genomic instability and smoking-associated mutagenesis.

COAD showed APC as the most frequently mutated gene followed by TP53, KRAS and PIK3CA, highlighting the importance of canonical colorectal cancer driver pathways.

Thus, LUAD is characterized mainly by mutations in large structural genes, whereas COAD is driven predominantly by pathway-specific oncogenic mutations.

4. SBS96 Decomposition Plot Comparison

The SBS96 decomposition plots compare extracted de novo mutational signatures with COSMIC reference signatures.

LUAD contained three extracted signatures and was dominated by SBS4 (tobacco smoking), together with SBS1, SBS5, SBS2, SBS13 and SBS6. This indicates contributions from smoking, ageing, APOBEC activity and limited mismatch repair deficiency.

COAD contained two extracted signatures composed mainly of SBS1, SBS5, SBS4, SBS10a, SBS10b, SBS13 and SBS15. These signatures indicate ageing, DNA polymerase proofreading defects, APOBEC activity and mismatch repair deficiency.

Overall, LUAD showed greater signature diversity, whereas COAD exhibited signatures closely associated with DNA repair defects.

5. COSMIC Signature Comparison

COSMIC Signature	LUAD	COAD	Biological Process / Interpretation
SBS1	Present	Present	Age-related spontaneous deamination of 5-methylcytosine (clock-like signature).
SBS2	Present	Present	APOBEC cytidine deaminase activity causing cytosine deamination.
SBS4	Major signature	Present (lower contribution)	Tobacco smoke-associated DNA damage.
SBS5	Present	Present	Clock-like signature that accumulates with age; ubiquitous across cancers.
SBS6	Present	Absent	DNA mismatch repair (MMR) deficiency.
SBS10a	Absent	Present	Defective DNA polymerase ϵ (POLE) proofreading.
SBS10b	Absent	Present	Defective DNA polymerase ϵ (POLE) proofreading.
SBS13	Present	Present	APOBEC-mediated mutagenesis; often accompanies SBS2.
SBS15	Absent	Present	Mismatch repair deficiency and microsatellite instability (MSI).

6. Signature Activity Plot Comparison

Signature activity plots display the contribution of each COSMIC signature across individual tumour samples.

LUAD showed marked inter-patient variability, with SBS4 dominating many samples while SBS1 and SBS5 were consistently present. SBS2, SBS13 and SBS6 were observed only in subsets of patients, reflecting molecular heterogeneity.

COAD demonstrated more uniform signature activities across samples. SBS1 and SBS5 were consistently active, whereas SBS15 and SBS10 signatures were enriched in selected tumours, indicating mismatch repair deficiency and polymerase proofreading defects.

Overall, LUAD exhibited greater patient-to-patient heterogeneity, whereas COAD displayed a more conserved mutational signature profile.

Overall Comparative Conclusion

Both LUAD and COAD are characterized by predominantly missense mutations and share common age-related mutational signatures (SBS1 and SBS5). However, LUAD is distinguished by smoking-associated mutagenesis (SBS4) and higher molecular heterogeneity, whereas COAD is characterized by APC-driven tumorigenesis, mismatch repair deficiency (SBS15) and DNA polymerase proofreading defects (SBS10a/SBS10b). These observations demonstrate that although both cancers accumulate somatic mutations, the biological mechanisms responsible for tumour development differ substantially.

Biological Interpretation

The biological interpretation summarizes the major findings obtained from mutation profiling and mutational signature analysis of Lung Adenocarcinoma (LUAD) and Colorectal Adenocarcinoma (COAD).

1. Mutation Summary

Both LUAD and COAD were dominated by missense mutations and SNPs, indicating that point mutations are the major genomic alterations. LUAD showed a higher mutation burden and greater sample-to-sample variability, whereas COAD displayed a more uniform mutation profile. C>A substitutions were more common in LUAD, consistent with tobacco-related DNA damage, while C>T substitutions predominated in COAD, reflecting age-related mutational processes.

Feature	LUAD	COAD
Dominant mutation	Missense, SNP	Missense, SNP
Major SNV	C>A	C>T
Mutation burden	Higher	Moderate
Heterogeneity	High	Moderate

2. Driver Genes

LUAD was characterized by frequent mutations in TP53, TTN and MUC16, suggesting genomic instability and smoking-associated mutagenesis. COAD was dominated by APC, TP53, KRAS and PIK3CA, highlighting disruption of the Wnt, MAPK and PI3K signalling pathways.

Feature	LUAD	COAD
Top genes	TP53, TTN, MUC16	APC, TP53, KRAS
Key pathway	Genome stability	Wnt/MAPK/PI3K

3. Mutational Signatures

Both cancers shared the age-related signatures SBS1 and SBS5. LUAD was dominated by SBS4, indicating tobacco-associated mutagenesis, together with APOBEC-associated SBS2 and SBS13. COAD showed SBS10a, SBS10b and SBS15, indicating defective DNA polymerase proofreading and mismatch repair deficiency.

Signature	LUAD	COAD	Interpretation
SBS1	✓	✓	Ageing

SBS4	Major	Minor	Tobacco exposure
SBS5	✓	✓	Clock-like
SBS6	✓	—	MMR deficiency
SBS10a/b	—	✓	POLE proofreading defect
SBS13	✓	✓	APOBEC activity
SBS15	—	✓	Mismatch repair deficiency

4. Signature Activity

LUAD exhibited greater inter-patient heterogeneity, with SBS4 contributing substantially in many samples. COAD showed more consistent activity of SBS1 and SBS5, while selected tumours demonstrated increased SBS15 and SBS10 activity, reflecting DNA repair defects.

Overall Biological Interpretation

Although LUAD and COAD share common age-related mutational processes, their dominant biological mechanisms differ. LUAD is primarily influenced by environmental factors, particularly tobacco exposure, resulting in a higher mutation burden and greater heterogeneity. In contrast, COAD is driven mainly by alterations in APC, KRAS and DNA repair pathways, including mismatch repair deficiency and defective DNA polymerase proofreading. These findings demonstrate distinct molecular mechanisms underlying the development of the two cancer types.

Conclusion

This project successfully compared the mutational landscapes of **Lung Adenocarcinoma (LUAD)** and **Colorectal Adenocarcinoma (COAD)** using publicly available TCGA somatic mutation data and open-source bioinformatics tools, including **maftools**, **SigProfilerMatrixGenerator**, and **SigProfilerExtractor**. Mutation profiling identified distinct driver genes, with **TP53**, **TTN**, and **MUC16** being frequently mutated in LUAD, while **APC**, **TP53**, **KRAS**, and **PIK3CA** predominated in COAD. Mutational signature analysis revealed that both cancers shared age-related signatures (SBS1 and SBS5), but differed in their dominant mutational processes, with LUAD showing a strong tobacco-associated signature (SBS4) and COAD exhibiting signatures linked to DNA mismatch repair deficiency and defective DNA polymerase proofreading (SBS15, SBS10a, and SBS10b). Overall, the study demonstrates how mutational signature analysis can uncover the biological mechanisms driving different cancer types and highlights the value of computational genomics in understanding cancer development, supporting biomarker discovery, and advancing precision oncology.

Software and Databases:

- **R (Version 4.5.1)**
- **maftools** (R package)
- **Python 3.12**
- **SigProfilerMatrixGenerator**
- **SigProfilerExtractor**
- **TCGA Genomic Data Commons (GDC)**
- **COSMIC v3.4 SBS Signatures**
- **GitHub**